

Agent Companies

Haotian Zhai

haotian.zhai@utexas.edu

Introduction

- What Are 'Agent Companies'?
 - Organizations built around AI agents as core workers
 - Agents perform planning, execution, and collaboration tasks
 - Human oversight remains, but coordination is shifting from *human-centered* to ***AI-centered***
- Why 'Agent Companies'?
 - Macro View — Global & Enterprise AI Diffusion
 - Meso View — Benchmarking AI Workplaces
 - Micro View — Human-Agent Collaboration

Macro View

The Anthropic Economic Index report: Uneven geographic and enterprise AI adoption

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Authors:

Ruth Appel*, Peter McCrory*, Alex Tamkin*
Miles McCain, Tyler Neylon, Michael Stern

Overview

- How [Claude.ai](#) is being used:
 - Over Time
 - Across World/US
 - Among Enterprise

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Claude: Usage Over Time

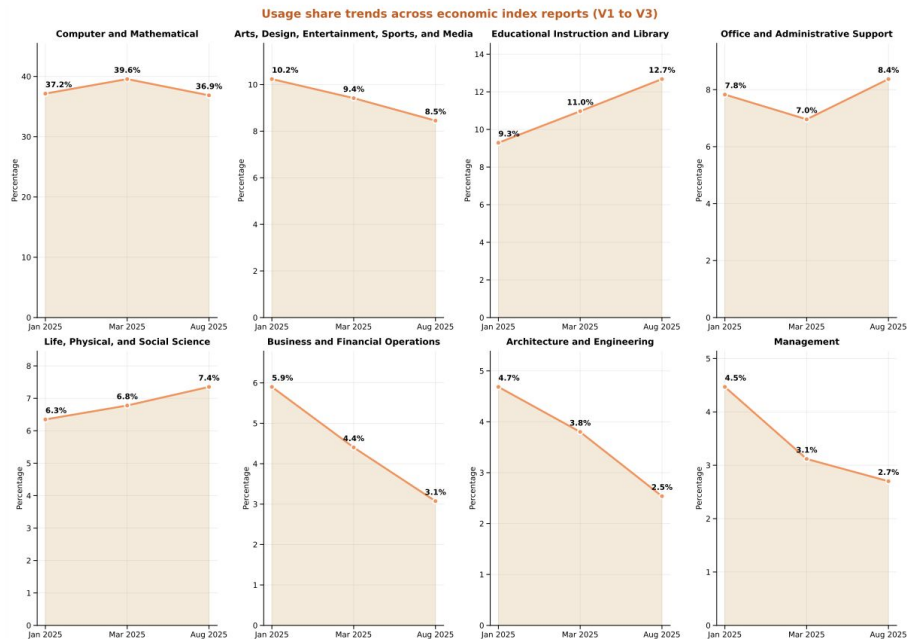


Figure 1.1: Claude.ai usage over time Each panel shows the share of sampled conversations on Claude.ai associated with tasks from each SOC major group. We see notable increases in usage for scientific and educational tasks. SOC major groups ranked by usage in our first report.

- Computer and Mathematical tasks remain the main part
- Educational and scientific tasks continue their rise in relative importance

Claude: Usage Over Time

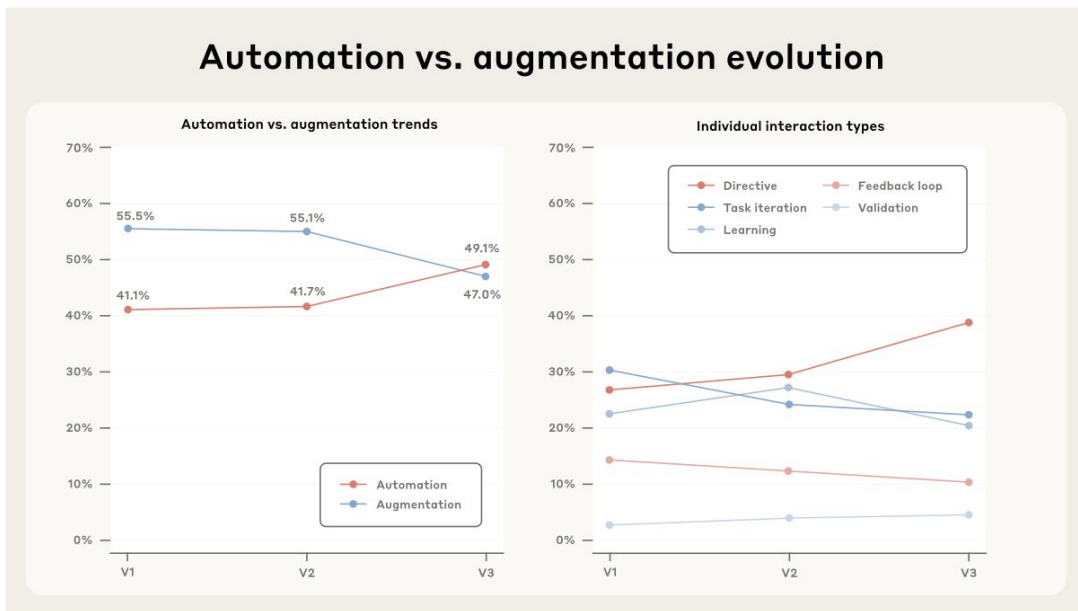


Figure 1.2: Collaboration mode frequencies across Anthropic Economic Index Reports The left panel calculates the share of conversations exhibiting either automation or augmentation forms of use. The right panel breaks this out by collaboration mode. Claude tends to be used in more automated ways over time, driven primarily by an increase in directive use.

- Automation:
 - Directive
 - Feedback loop
- Augmentation
 - Task Iterations
 - Validation
 - Learning

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Claude: Usage Across World

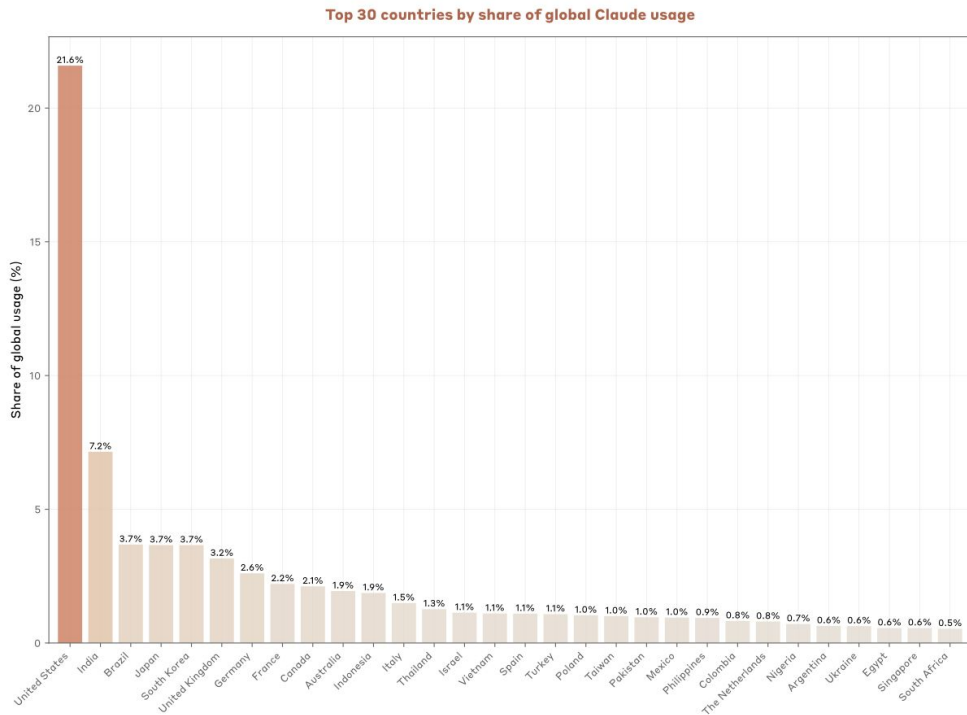
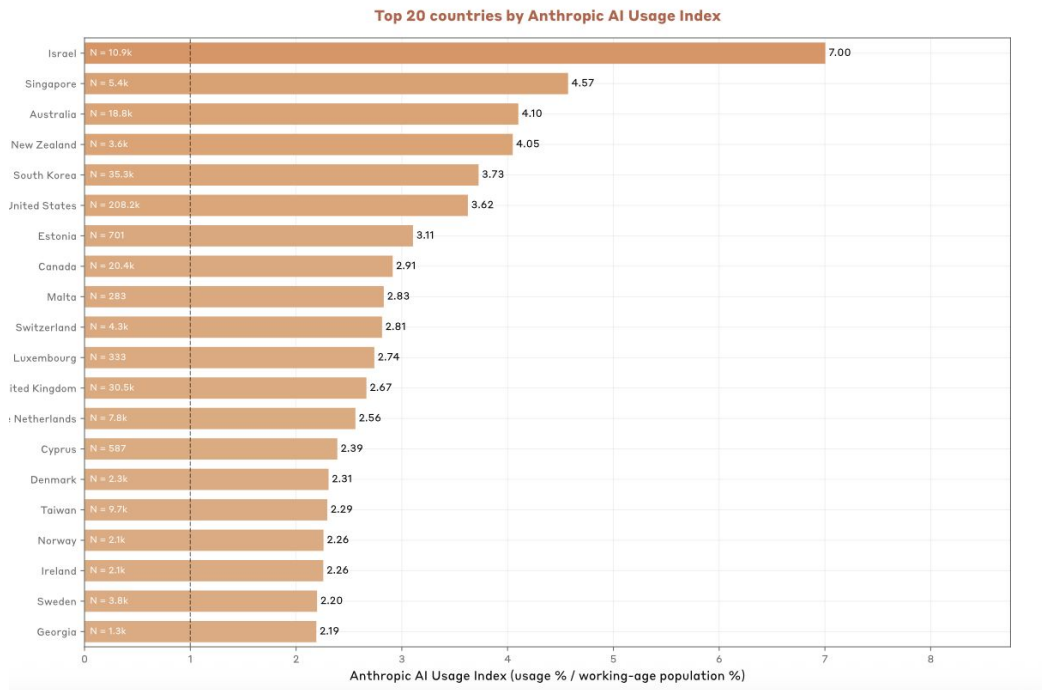


Figure 2.1: Leading countries in terms of global Claude.ai usage share The data includes Claude.ai Free and Pro conversations.

Claude: Usage Across World

AI Usage Index (AUI): For each geography, we calculate its share of Claude usage, and its share of the working-age population (ages 15-64). We then calculate the AUI by dividing these shares:

Claude: Usage Across World



Claude: Usage Across World

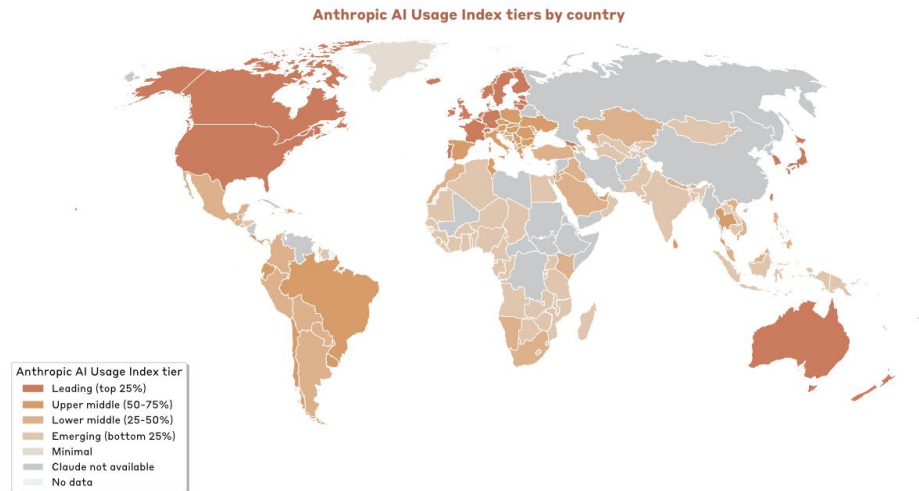


Figure 2.3: Claude diffusion varies across countries, with countries in North America, Europe and Oceania leading in Claude adoption per working-age capita The different tiers reflect a country's position within the global distribution of the Anthropic AI Usage Index as defined in this chapter.^{5,6}

Claude: Usage Across World

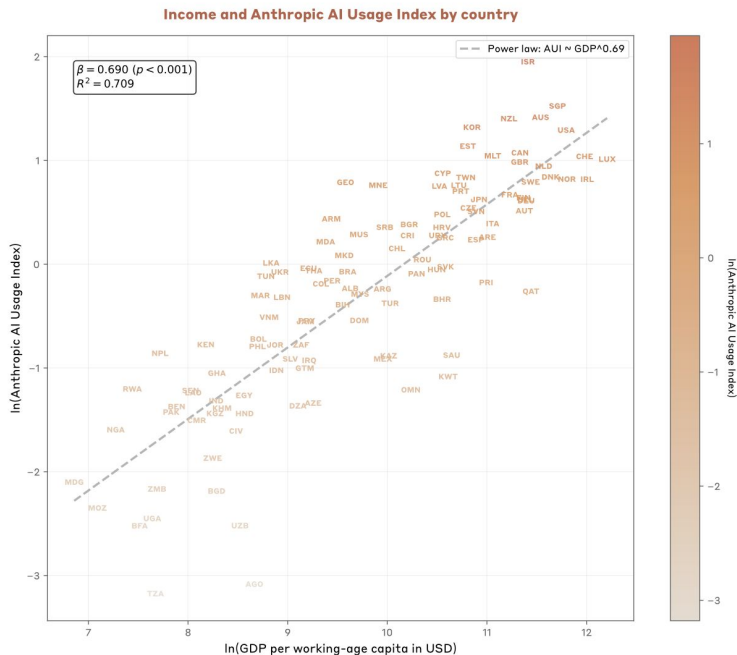


Figure 2.4: Claude usage per capita is positively correlated with income per capita across countries. We only include countries with at least 200 observations in our sample for this figure because of the uncertainty of the measure for low-usage countries in our random sample. Axes are on a log scale, highlighting a power law distribution. Each country is represented by its 3-letter ISO code.

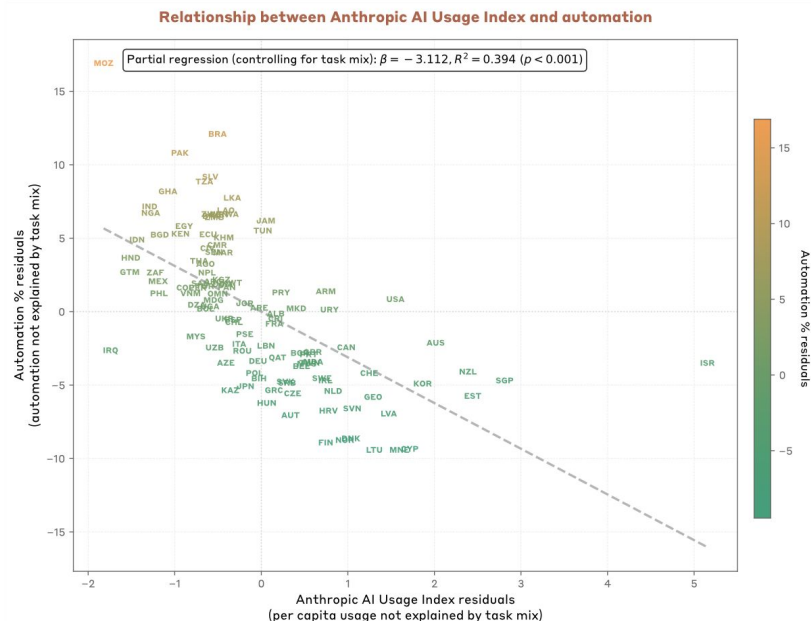


Figure 2.11: Countries with higher Anthropic AI Usage Index tend to use Claude in a more collaborative manner (augmentation), rather than have it operate independently (automation). This figure shows the relationship between the Anthropic AI Usage Index and the automation share in a given country. We plot the relationship after accounting for a geography's task mix, thus we show the regression residuals. We only include countries with at least 200 observations in our sample for this figure because of the uncertainty of the measure for low-usage countries in our random sample. Each country is represented by its 3-letter ISO code.

Claude: Usage Across World

Top overrepresented requests for the United States, Brazil, Vietnam and India



Figure 2.8: Overrepresented request clusters for the United States, Brazil, Vietnam and India A request is overrepresented in a country when the share of conversations containing that request is higher for that country than globally. For this figure, we focus on request clusters at the middle level of granularity, i.e. more aggregated than the lowest level request clusters, but less aggregated than the highest level request clusters. Only includes requests with at least 1% frequency globally and for that country.

Claude: Usage Across US

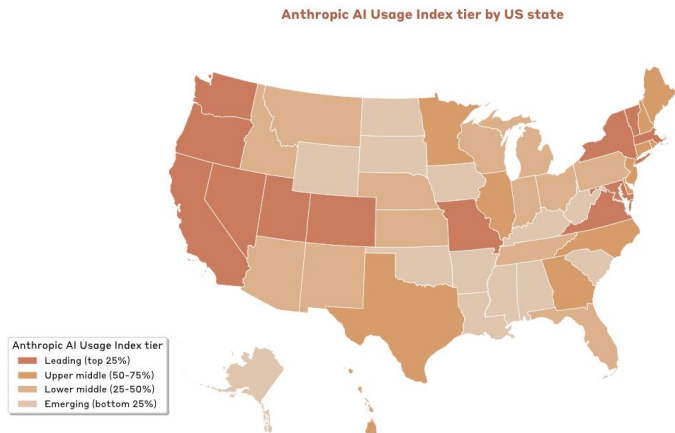


Figure 2.6: Claude usage varies across US states, with high per-capita usage in the West Coast, but also higher usage in Nevada, Utah, Colorado, Missouri, and Virginia The different tiers reflect a US state's position within the US distribution of the Anthropic AI Usage Index as defined in this chapter.

Tier	AUI range	# of states	Example states
Leading (top 25%)	0.98 - 3.82	13	District of Columbia, Utah, California, New York, Virginia
Upper middle (50-75%)	0.71 - 0.88	12	Illinois, Connecticut, New Hampshire, New Jersey, Hawaii
Lower middle (25-50%)	0.43 - 0.70	13	Arizona, Florida, Pennsylvania, Tennessee, New Mexico
Emerging (bottom 25%)	0.21 - 0.42	13	South Carolina, Alabama, Wyoming, North Dakota, Iowa

Table 2.2: Claude per capita usage tiers with examples, number of states, and AUI range for each tier.

Claude: Usage Across US

Top overrepresented high-level requests for California, Texas, Florida and South Carolina

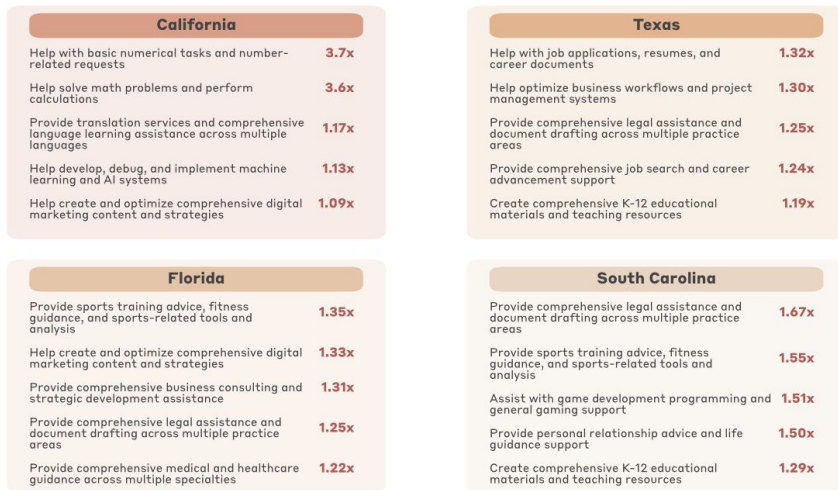


Figure 2.9: Overrepresented request categories for California, Texas, Florida and South Carolina A request is overrepresented in a state when the share of conversations containing that request is higher for that state than in the US as a whole. For this figure, we focus on request clusters at the middle level of granularity, i.e. more aggregated than the lowest level request clusters, but less aggregated than the highest level request clusters. Only includes requests with at least 1% frequency in the United States and for that state.

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Claude: Usage Among Enterprise

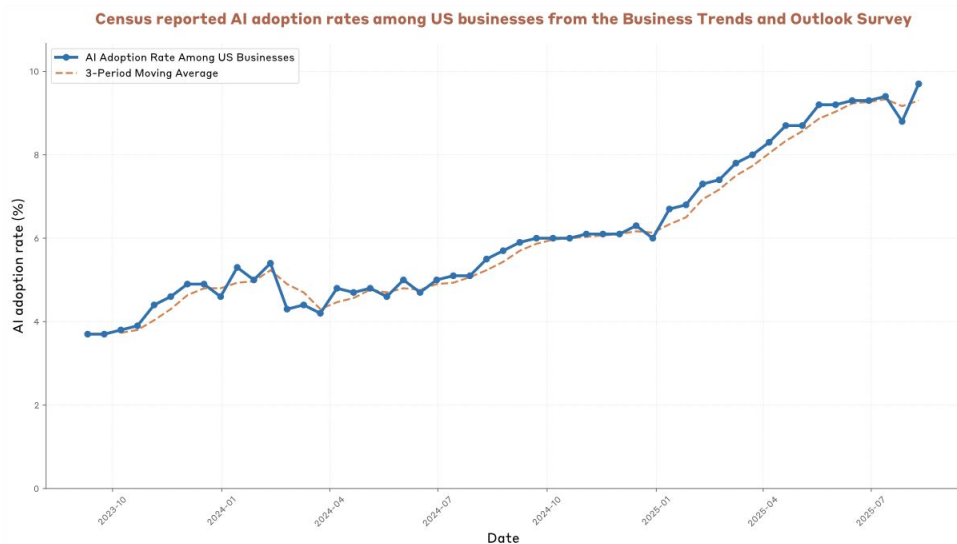


Figure 3.1: AI adoption rates among US firms, Business Trends & Outlook Survey (Census) Note: AI adoption rates are calculated as the share of firms responding “yes” to the question “In the last two weeks, did this business use Artificial Intelligence (AI) in producing goods or services? (Examples of AI: machine learning, natural language processing, virtual agents, voice recognition, etc.)”.

Claude: Usage Among Enterprise

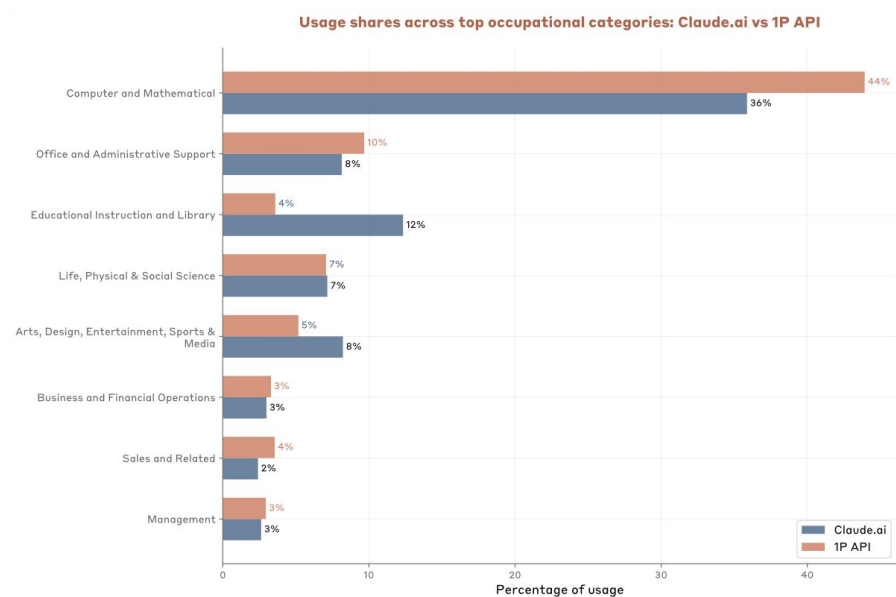


Figure 3.3: Leading Occupational Categories by Overall Usage: Claude.ai vs 1P API After determining usage shares for tasks, we calculate the share of traffic from Claude.ai and 1P API customers assigned to top-level occupations in the O*NET taxonomy. For example, this figure shows that 44% of API traffic in our sample was matched to a task characteristic of a Computer and Mathematical occupation.

From Global Diffusion to Workplace Simulation

- Global usage patterns reveal massive interest
 - Increasing usage patterns over time/among enterprises
- However, adoption is not equivalent to automation
 - The **Meso View** explores what AI can actually *do* in real workplaces

Meso View

Remote Labor Index: Measuring AI Automation of Remote Work

Mantas Mazeika^{*1}, Alice Gatti^{*1}, Cristina Menghini^{*†},
Udari Madhushani Schwag^{*2}, Shivam Singhal^{*†}, Yury Orlovskiy^{*1}

Steven Basart¹, Manasi Sharma², Denis Peskoff², Elaine Lau², Jaehyuk Lim¹,
Lachlan Carroll¹, Alice Blair¹, Vinaya Sivakumar¹, Sumana Basu², Brad Kenstler²,
Yuntao Ma[†], Julian Michael[†], Xiaoke Li¹, Oliver Ingebrechtsen¹, Aditya Mehta¹,
Jean Mottola¹, John Teichmann[†], Kevin Yu[†], Zaina Shaik[†], Adam Khoja¹,
Richard Ren¹, Jason Hausenloy¹, Long Phan¹, Ye Htet², Ankit Aich²,
Tahseen Rabbani², Vivswan Shah[†], Andriy Novykov¹, Felix Binder[†]

Kirill Chugunov², Luis Ramirez², Matias Geralnik², Hernán Mesura²,
Dean Lee[†], Ed-Yeremai Hernandez Cardona², Annette Diamond[†]

Summer Yue^{**†}, Alexandr Wang^{**†},
Bing Liu^{**2}, Ernesto Hernandez^{**2}, Dan Hendrycks^{**1}

¹Center for AI Safety ²Scale AI

RLI: Motivation

1. **Benchmark gap:** AI excels on academic tests (MMLU, GPQA, etc.) but we don't know how much real, paid human work it can actually do.
2. **Economic need:** No empirical way to measure AI's impact on real-world labor and automation — RLI provides that foundation.

RLI: Example

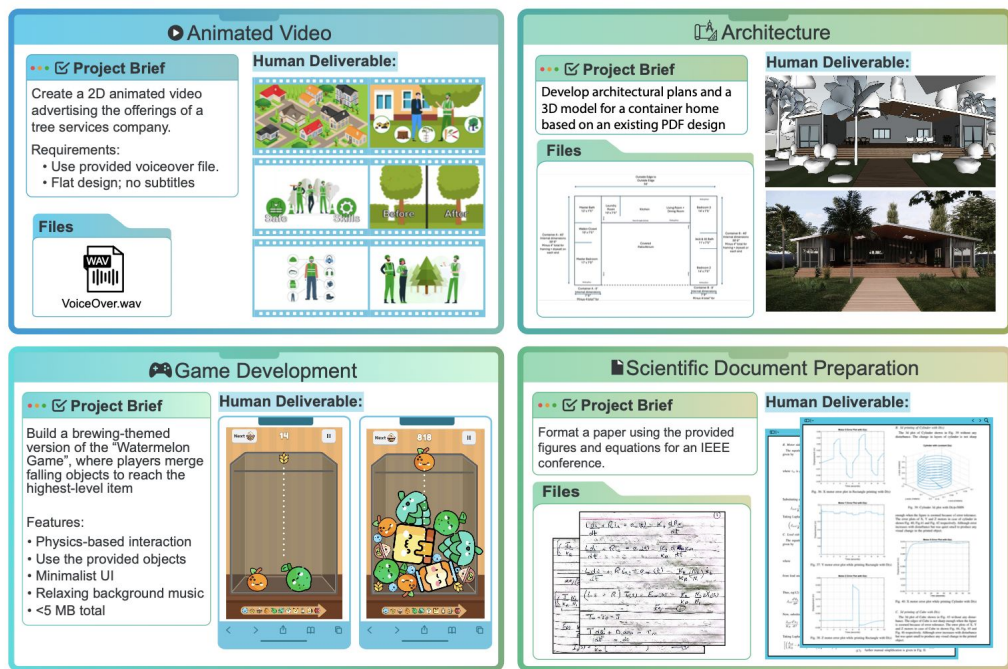


Figure 1: The Remote Labor Index (RLI) represents a broad range of projects from across the remote labor economy, including game development, product design, architecture, and data analysis. All projects represent real work that was performed by human professionals.

RLI: Components

Project composition. Each project in RLI consists of three components:

- **Brief:** A text document describing the work to be done
- **Input files:** A directory containing files needed to complete the project
- **Human deliverable:** A gold-standard deliverable that successfully completes the project, produced by a professional

RLI: Data Collection

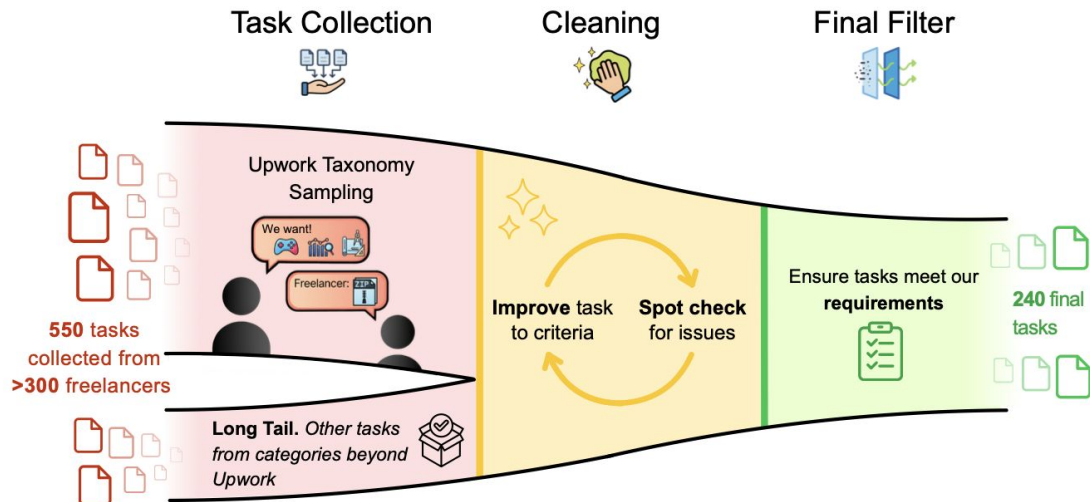


Figure 5: RLI projects were extensively filtered and cleaned to ensure quality. Projects were sourced primarily from the remote labor market and secondarily from deliverables representing uncommon and emerging types of remote work work. (For details, see Appendix [C](#).)

RLI: Complexity

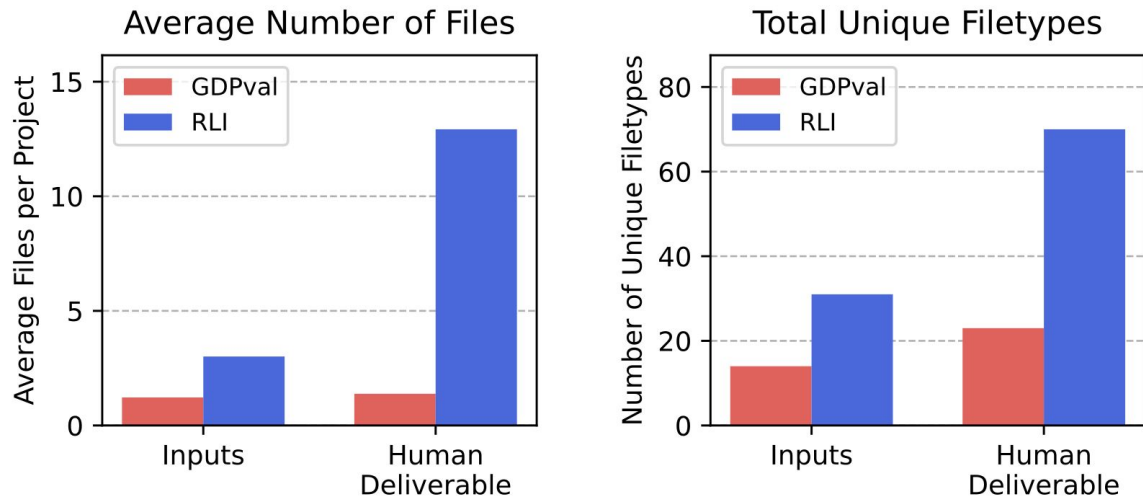


Figure 14: RLI projects involve significantly more diverse file types than previous comparable benchmarks. Left: Average number of files per project for inputs and human deliverables across benchmarks. Right: Total unique file types found in inputs and human deliverables across benchmarks.

RLI: Complexity

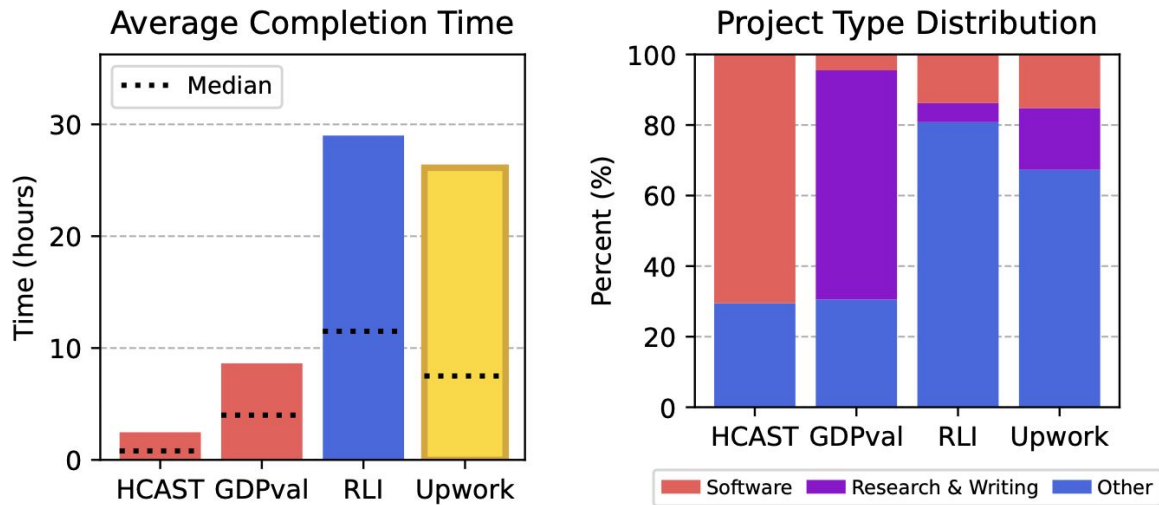


Figure 6: RLI is far closer to the complexity and diversity of real freelance labor than previous comparable benchmarks. Left: The average completion time for humans on RLI projects matches the true Upwork distribution. Right: Previous benchmarks primarily focus on tasks involving software engineering or web-based research and writing, but real remote labor markets have far more diversity.

RLI: Complexity

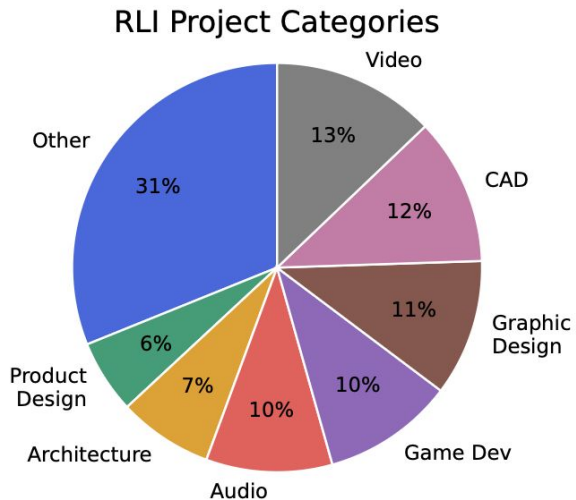


Figure 3: RLI captures a wide array of project types, spanning 23 categories of work from the Upwork taxonomy. Here, we show the top seven categories.

RLI: Complexity

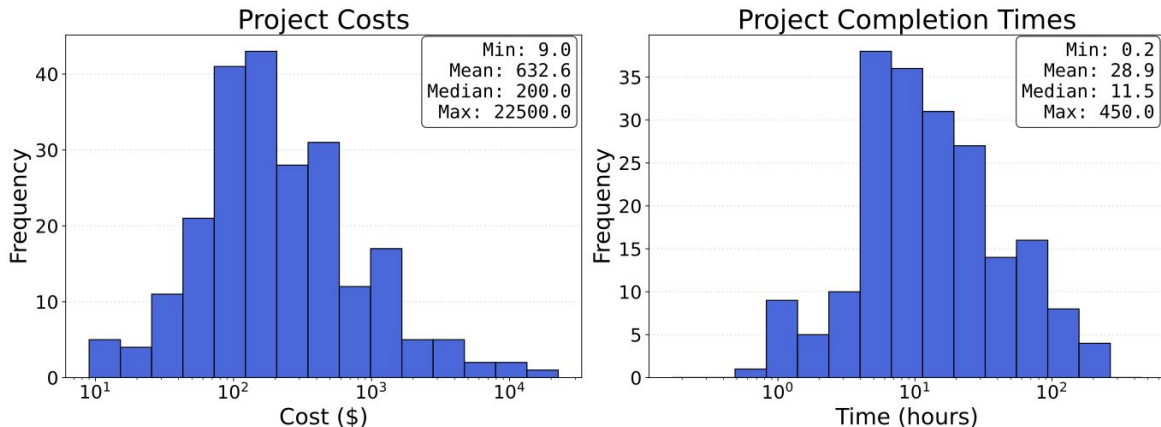


Figure 4: RLI spans a broad range of difficulty, with project costs reaching over \$10,000 and completion times for human professionals reaching over 100 hours. All project costs and completion times come directly from human professionals who completed the projects. In total, the projects in RLI represent over 6,000 hours of real work valued at over \$140,000.

RLI: Metrics

Automation rate: The percentage of projects for which the AI deliverable is judged by human evaluators to complete the project at least as well as the human deliverable. This measures the absolute success rate of the AI agent across RLI projects.

Elo: A score capturing the relative performance of different AI agents. For each project, a deliverable from two different AIs is presented to human evaluators, who judge which deliverable is closer to completing the project successfully. If both agents successfully complete the project, then their deliverables are compared on overall quality. A difference of 400 corresponds to 10:1 odds of winning.

RLI: Evaluation

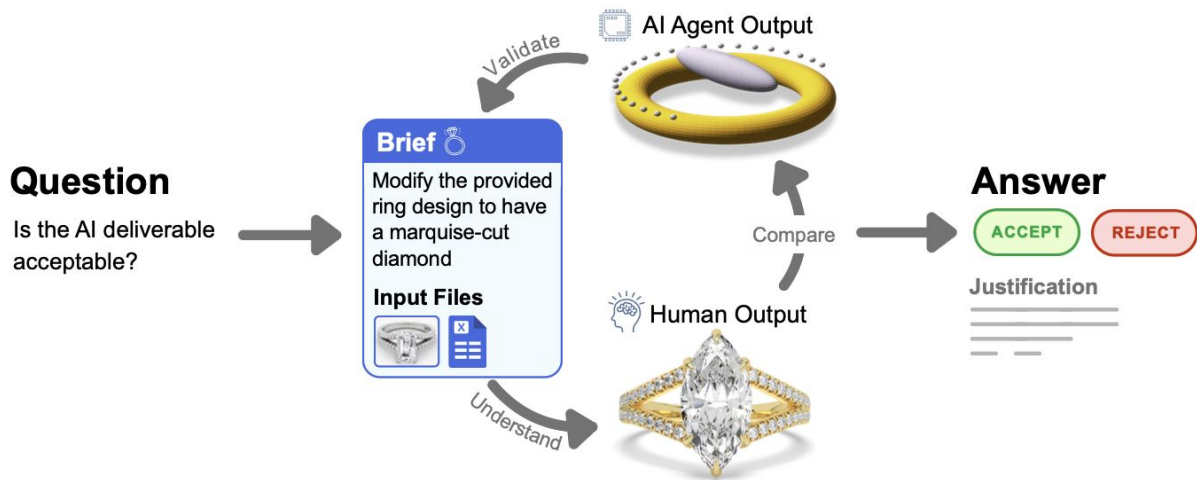


Figure 7: Evaluation Pipeline: For each RLI project, AI deliverables are rigorously checked against human gold-standard deliverables and the requirements in the project brief for flaws and to determine whether the AI deliverable would be accepted as work product in a realistic freelance setting. Evaluating AI deliverables is itself a highly agentic task, so automating evaluation with LLMs is not currently feasible. Thus, all evaluations are performed manually by trained workers and subject experts. Inter-annotator agreement is above 94%.

RLI: Results

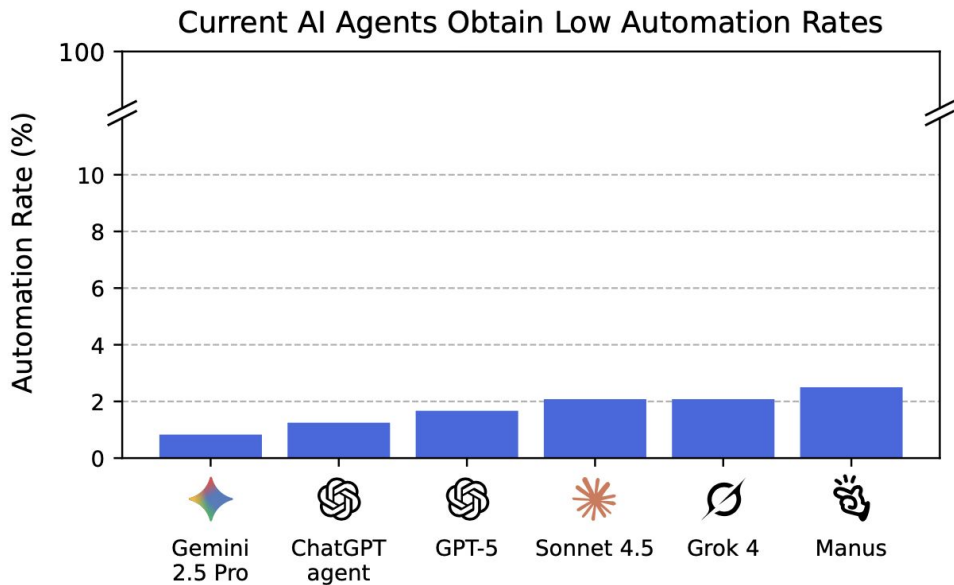


Figure 2: All AI agents tested automate at most 2.5% of tasks on RLI, showing that most economically valuable remote work currently remains far beyond their capabilities.

RLI: Results

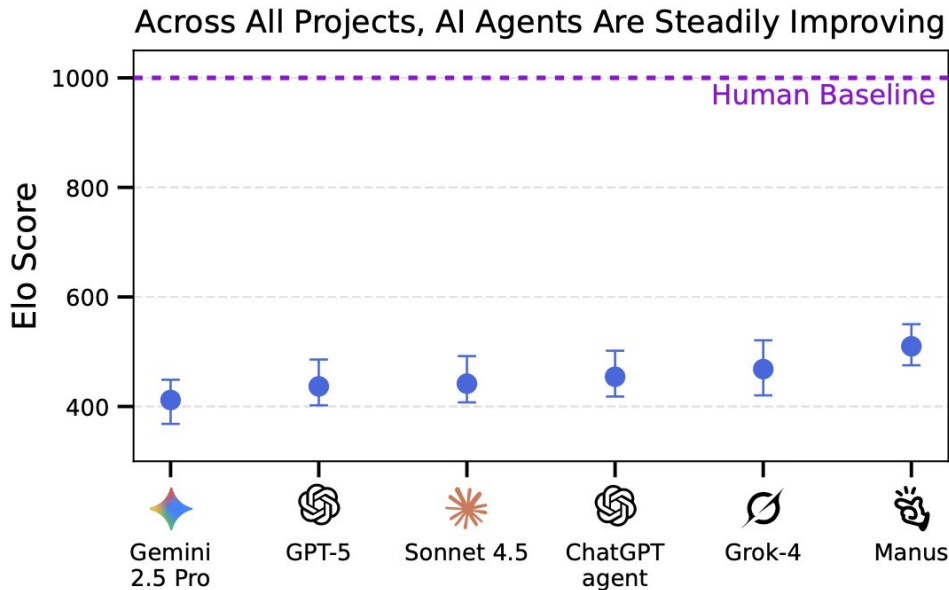


Figure 8: Relative performance (Elo) scores show that AI agents are making steady progress on RLI and there are meaningful differences between models, despite all models falling short of the human baseline of 1,000. Compared to the automation rate metric, Elo score provides a better measure of partial progress across all projects, including projects that are not solved yet.

RLI: Results

Common failure modes. Our qualitative analysis across roughly 400 evaluations shows that rejections predominantly cluster around the following primary categories of failure:

1. **Technical and File Integrity Issues:** Many failures were due to basic technical problems, such as producing corrupt or empty files, or delivering work in incorrect or unusable formats.
2. **Incomplete or Malformed Deliverables:** Agents frequently submitted incomplete work, characterized by missing components, truncated videos, or absent source assets.
3. **Quality Issues:** Even when agents produce a complete deliverable, the quality of the work is frequently poor and does not meet professional standards.
4. **Inconsistencies:** Especially when using AI generation tools, the AI work often shows inconsistencies between deliverable files.

Meso View

TheAgentCompany: Benchmarking LLM Agents on Consequential Real World Tasks

Frank F. Xu^{1*} Yufan Song^{2*} Boxuan Li^{2*} Yuxuan Tang² Kritanjali Jain¹
Mengxue Bao² Zora Z. Wang¹ Xuhui Zhou¹ Zhitong Guo¹ Murong Cao²
Mingyang Yang² Hao Yang Lu² Amaad Martin¹ Zhe Su¹ Leander Melroy Maben¹
Raj Mehta¹ Wayne Chi¹ Lawrence Jang¹ Yiqing Xie¹ Shuyan Zhou³ Graham Neubig¹
¹Carnegie Mellon University ²Independent ³Duke University
{fangzhex, gneubig}@cs.cmu.edu, {yufans, boxuanli}@alumni.cmu.edu

TheAgentCompany: Overview

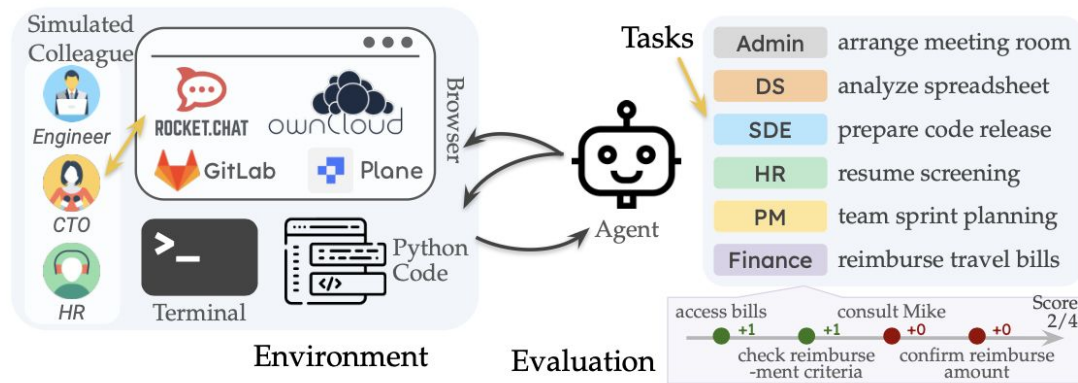


Figure 1: An overview of TheAgentCompany benchmark. It features a reproducible and self-hosted environment, simulated colleagues to test agent communication capabilities, checkpoint and execution-based evaluation, and a set of 175 diverse, realistic and professional tasks in a software engineering company setting.

TheAgentCompany: Comparison

Table 2: Comparison of different AI agent benchmarks. **Interface**: the interface agent has access to;  is web browser,  is desktop,  is API usage,  is Python runtime,  is chat platform,  is bash terminal. **Task Categories**: tasks in the benchmark, * indicate tasks with no association with real-world occupations; SE refers to software engineering, HR is human resources, PM is project management. **Long-Horizon with Checkpoints**: if tasks are evaluated at intermediate checkpoints and assigned partial scores. **Requires Interaction**: If the agent can interact with other NPC agents during task-solving.

Framework	Diverse Real-world Work	Task Categories	Requires Interaction	Long-Horizon w/ Checkpoints	Interface	Self-Hosted Environment
MiniWob++ (Liu et al., 2018)	✗	Browsing*	✗	✗		✓
Mind2Web (Deng et al., 2023)	✗	Browsing*	✗	✗		✗
WebLINX (Lü et al., 2024)	✗	Browsing*	✗	✗		✗
AssistantBench (Yoran et al., 2024)	✗	Browsing*	✗	✗		✗
WebArena (Zhou et al., 2023)	✗	Browsing*	✗	✗		✓
VisualWebArena (Koh et al., 2024)	✗	Browsing*	✗	✗		✓
VideoWebArena (Jang et al., 2024)	✗	Browsing*	✗	✗		✓
WorkArena (Drouin et al., 2024)	✓	Enterprise Software	✗	✗		✗
OSWorld (Xie et al., 2024)	✓	Office, Coding	✗	✗		✓
Windows Agent Arena (Bonatti et al., 2024)	✓	Browsing*, Office, Coding	✗	✗		✓
CRMArena Huang et al. (2024)	✓	Service agent, Analyst, Manager	✗	✗		✗
AppWorld (Trivedi et al., 2024)	✗	Daily	✗	✗		✓
Gorilla APIBench (Patil et al., 2023)	✗	Coding	✗	✗		✓
τ -bench (Yao et al., 2024)	✓	Retail, Airline	✓	✗		✗
SWE-bench (Jimenez et al., 2024)	✗	SWE	✗	✗		✓
DevBench (Li et al., 2024)	✗	SWE	✗	✗		✗
Smallville (Park et al., 2023)	✗	Social*	✓	✗		✓
Sotopia (Zhou et al., 2024)	✗	Social*	✓	✗		✓
TheAgentCompany	✓	SWE, HR, Admin, PM, Research, Finance	✓	✓	   	✓

TheAgentCompany: Simulated Colleagues

Emily Zhou
Female, 29 years old
Software Engineer (Web Crawler Team)

 Good morning! I'd like to schedule a meeting with you and Liu. When are you available? 9:43 AM

 Hi there, I'm available for a meeting on Wednesday or Thursday. What time works best for you on those days? 10:03 AM

 Great! I'm also available on Wednesday afternoons or Thursday mornings. Let me check Liu Qiang's schedule and confirm. 10:04 AM

 Sounds good. Once you've checked with Liu, let me know the final time for Wednesday or Thursday, and I'll block it on my calendar. 10:04 AM

 Hello Emily! The final meeting time is 10:30 AM on Thursday. 10:38 AM

Liu Qiang
Male, 36 years old
Quality Assurance Engineer

 Hi Liu Qiang, I'd like to schedule a meeting with you and Emily. She's available on Wednesday or Thursday. When are you available? 10:05 AM

 Hello! I'm available on Thursday for the meeting with Emily. Please let me know what time works best for both of you. 10:24 AM

 Oh good, I think Thursday works for us. Could we aim for a mid-morning slot? Perhaps around 10:30 AM? 10:25 AM

 Yup. That works for me. I will send Emily a link myself. 10:37 AM

 Thank you. The final meeting time is 10:30 AM on Thursday. Have a great day! 10:38 AM

- Simulated using Sotopia, which supports the creation of simulated human characters with LLMs.
- Each simulated colleague is equipped with a detailed profile that includes their name, role, responsibilities, and project affiliations.

Figure 7: Simulated Colleague Communication Example 3 - The agent is tasked with scheduling a meeting between NPCs Emily Zhou and Liu Qiang based on their availability. Emily is available on Wednesday and Thursday, while Liu is only available on Thursday. The agent identifies Thursday as the common free day and successfully proposes a mid-morning slot at 10:30 AM, which both participants confirm. This example highlights the agent's ability to manage multi-turn conversations, effectively going back and forth between participants to align schedules and finalize a meeting time.

TheAgentCompany: Task Structure

Table 3: Example task intents and checkpoints for three domains.

Domain	Task Intent	Checkpoints
SWE	Set up JanusGraph and run it locally with an HTTP endpoint: - Clone JanusGraph directory under <code>/workspace</code> folder (http://the-agent-company.com:8929/root/janusgraph). - Build the binary file. - Launch JanusGraph server locally on port 8182 with an HTTP endpoint.	Checkpoint 1 (1pt): Check if JanusGraph repo is cloned. Checkpoint 2 (3pts): Check if the binary file is built (requires skipping Docker in <code>pom.xml</code> , hence higher points). Checkpoint 3 (2pts): Check the JanusGraph Server as an HTTP endpoint.
Finance	Navigate to ownCloud at http://the-agent-company.com:8092 and complete Section B—Alternative Simplified Credit of IRS Form 6765: - Gather necessary information from <code>/Documents/Financials/TAC_financials.csv</code> and <code>/workspace/research_wages.csv</code>. - Consult <code>/Documents/Financials/f6765_instructions.pdf</code> for instructions. - Contact the finance director (David Wong) on Rocket.Chat (http://the-agent-company.com:3000/home) for ambiguous questions. - Save the filled form as <code>/workspace/filled_f6765.pdf</code>.	Checkpoint 1 (5pts): Check if all 16 questions in Section B of the form have been answered correctly. Checkpoint 2 (3pts): Check if the correct finance director (David Wong) was contacted to answer two ambiguous questions.
PM	Analyze The Agent Company's performance and create a summary in Plane: - Access Plane (http://the-agent-company.com:8091/tac/) and navigate to "Analytics." - Collect metrics: Open Tasks, Backlog Tasks, Unstarted Tasks, Started Tasks, Unassigned Issues, Pending Issues. - Create a summary and share it on Rocket.Chat (http://the-agent-company.com:3000/home) in the <code>#kudos</code> channel.	Checkpoint 1 (1pt): Check if Plane was accessed and the agent navigated to "Analytics" section. Checkpoint 2 (3pts): Check if all required project metrics were collected. Checkpoint 3 (1pt): Check if the summary was shared in the <code>#kudos</code> channel on Rocket.Chat.

- Tasks have multiple checkpoints (Action, Data Accuracy, Collaboration), measuring intermediate milestones.

TheAgentCompany: Example Task

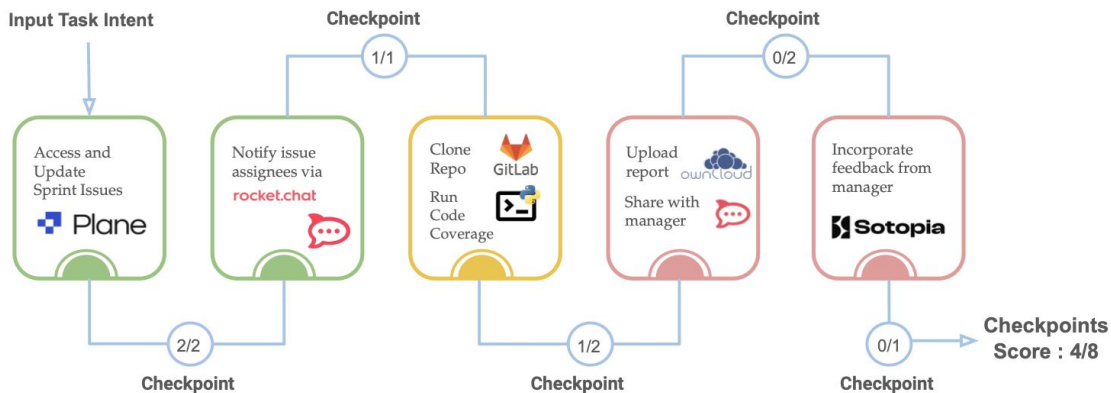


Figure 3: Example TheAgentCompany workflow illustrating an agent managing a sprint for the *RisingWave* project. The task involves identifying and moving unfinished issues to next sprint cycle, notifying assignees of those issues, running a code coverage script, uploading summarized report to OwnCloud, and incorporating feedback on report from a simulated project manager.

TheAgentCompany: Task Curation

1. Create a description of task intent, checkpoints, and how to evaluate each checkpoint.
2. Identify and import the required data for the task that are currently missing in the company Intranet services and create any necessary data.
3. Write scripts to configure the required initialization state in the local workspace.
4. Implement the checkpoint evaluators that calculate the scalar scores for each checkpoint.

TheAgentCompany: Evaluation

Full completion score We define the **full completion score** S_{full} as:

$$S_{\text{full}} = \begin{cases} 1 & \text{if all checkpoints are successfully passed,} \\ 0 & \text{otherwise.} \end{cases}$$

This binary metric evaluates if the agent successfully completed the task by passing all checkpoints.

Partial completion score To provide a more nuanced measure that rewards partial task completion while strongly incentivizing full task completion, we define **partial completion score** as: $S_{\text{partial}} = 0.5 \cdot \frac{\text{Result}}{\text{Total}} + 0.5 \cdot S_{\text{full}}$, where: “Result” is sum of awarded points across all checkpoints (including partial credit), “Total” is sum of the total points for all checkpoints, $\frac{\text{Result}}{\text{Total}}$ is fractional progress toward full completion, and S_{full} is binary indicator equal to 1 when the task is fully completed.

TheAgentCompany: Results

Table 1: Performance comparison of various foundation models on TheAgentCompany.

Agent	Model	Success	Score	Steps	Costs
<i>API-based Models</i>					
OpenHands 0.28.1	Gemini-2.5-Pro	30.3%	39.3%	27.2	\$4.2
OpenHands 0.28.1	Claude-3.7-Sonnet	26.3%	36.4%	27.8	\$4.1
OpenHands 0.14.2	Claude-3.5-Sonnet	24.0%	34.4%	29.2	\$6.3
OpenHands 0.14.2	Gemini-2.0-Flash	11.4%	19.0%	39.9	\$0.6
OpenHands 0.14.2	GPT-4o	8.6%	16.7%	14.6	\$1.3
OWL RolePlay	GPT-4o, o3-mini	4.0%	11.3%	N/A	N/A
OpenHands 0.14.2	Gemini-1.5-Pro	3.4%	8.0%	22.1	\$6.8
OpenHands 0.14.2	Amazon-Nova-Pro-v1	1.7%	5.7%	19.6	\$1.6
<i>Open-weights Models</i>					
OpenHands 0.14.2	Llama-3.1-405b	7.4%	14.1%	23.0	\$3.2
OpenHands 0.14.2	Llama-3.3-70b	6.9%	12.8%	20.9	\$0.9
OpenHands 0.14.2	Qwen-2.5-72b	5.7%	11.8%	24.0	\$1.5
OpenHands 0.14.2	Llama-3.1-70b	1.7%	6.5%	19.2	\$0.8
OpenHands 0.14.2	Qwen-2-72b	1.1%	4.2%	23.7	\$0.3

TheAgentCompany: Results

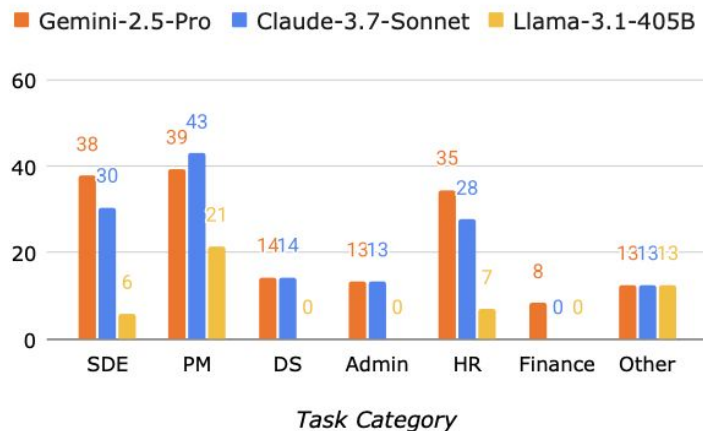
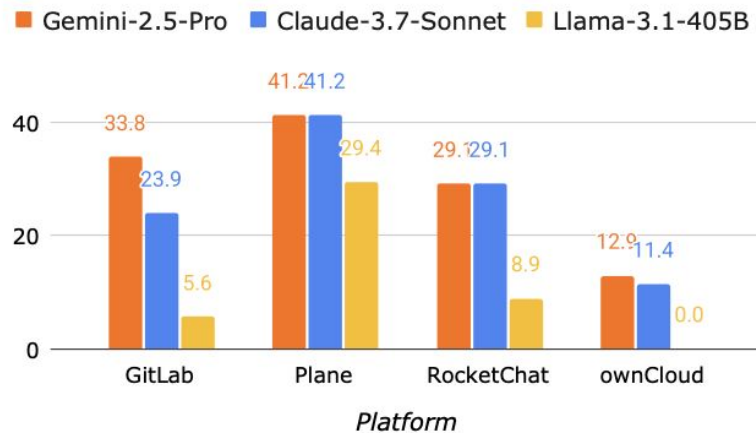


Figure 2: Comparing OpenHands success rate across platforms (left) and task categories (right).

TheAgentCompany: Results

Lack of social skills Sometimes, the agent fails to understand the implications and goals in the social conversations with colleagues in TheAgentCompany. For example, one task involves asking Alex for help, and the agent first successfully asks the right question “Could you tell me who I should introduce myself to next on the team?” Then the simulated colleague Alex replied “You should introduce yourself to Chen Xinyi next. She’s on our frontend team and would be a great person to connect with!” At this point, a human would then talk to Chen Xinyi, but instead the agent then decides to not follow up with her, and prematurely considers the task accomplished.

Incompetence in browsing Oftentimes, the biggest obstacle in tasks is the parts that require browsing the Web. This is expected as browsing is still hard for agents given the complexity of modern-day web UIs and the numerous distractions on a webpage. For example, on many tasks that involve ownCloud, a closable welcome popup has become an obstacle for OpenHands agent which uses text-based browsing. OpenHands agent gets stuck and fails to click on the ‘x’ to close the popup, while OWL RolePlay, which uses visual browsing, suffers less from this problem. On the other hand, OWL gets lost in complex web UIs more easily and clicks on wrong elements more often than OpenHands, although both agents share the same problem.

Deceiving oneself Interestingly, we find that for some tasks, when the agent is not clear what the next steps should be, it sometimes try to be clever and create fake “shortcuts” that omit the hard part of a task. For example, during the execution of one task, the agent cannot find the right person to ask questions on RocketChat. As a result, it then decides to create a shortcut solution by renaming another user to the name of the intended user.

From Multi-Agent Firms to Human-Agent Collab

- As agent firms scale, how should human adapt to it?
 - What is a better way of collaboration between humans and agents?

Micro View

Code with Me or for Me? How Increasing AI Automation Transforms Developer Workflows

Valerie Chen^{1,2}, Ameet Talwalkar¹, Robert Brennan², and Graham Neubig^{1,2}

¹Carnegie Mellon University

²All Hands AI

Correspondence to valeriechen@cmu.edu

Motivation

Benchmarks don't capture *how real developers* interact with coding agents.

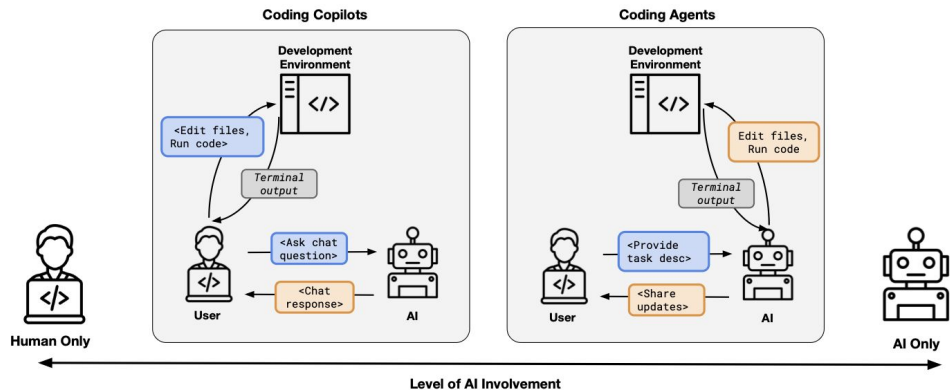


Figure 1: **The spectrum of developer workflows with AI tools.** While prior studies have evaluated how human-only workflows compare to human-copilot workflows, understanding how AI can code *with* developers, little work exists comparing copilots to the emerging agent workflows, where AI is more often used to code *for* developers. We study how these differences affect productivity, user experience, and interaction patterns with AI tools.

Research Questions

- RQ1:** *How do agents and copilots compare in terms of effect on user productivity?* While agents are often marketed as AI software engineers, we evaluate to what extent they can help a user complete more tasks than when a user has access to a widely used copilot tool.
- RQ2:** *How do agents and copilots compare in terms of effect on user experience?* Extensive surveys have been conducted to understand user experience with copilots. We ask many of the same questions that have been explored in prior surveys but directly comparing agents and copilots.
- RQ3:** *How do developers interact with different coding assistants?* Prior work has explored modeling user behaviors with copilots [21, 19, 18]. We compare interaction patterns between both tools.

Study Overview

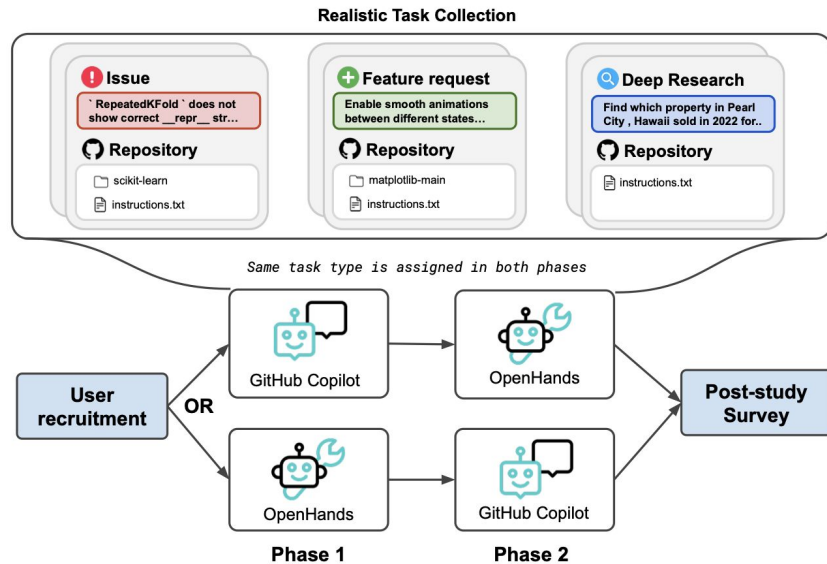


Figure 3: **Overview of our study procedure.** We designed a controlled user study to directly contrast participant interactions with copilots and agents. Participants were asked to solve a set of realistic and challenging tasks, including fixing issues in repositories, adding features, and conducting deep research and analysis.

Participants

Participants are selected as follows:

1. 20 students via university mailing lists
2. Have access to and use GitHub Copilot regularly and have experience programming in Python
3. 60% had 3-5 years of professional programming experience, 25% had 0-2 years, and 15% had 6-10 years
4. 95% of participants rated themselves as having intermediate to advanced proficiency in Python
5. All participants used GitHub Copilot at least once a week in addition to other common chat assistants like ChatGPT, Claude, and Gemini
6. No participants reported any prior experience with using coding agents

Tools Compared

Tool	Basic Features		Advanced Capabilities			
	Auto-complete	Chat	Multi-file Editing	Terminal Execution	File Creation	Act Autonomously
Copilots						
GitHub Copilot* [1]	✓	✓	✗	✓	✓	×
Cursor [2]	✓	✓	✗	✓	✓	×
Windsurf [3]	✓	✓	✗	×	×	×
Continue [4]	✓	✓	×	×	×	×
Amazon CodeWhisperer [49]	✓	✓	×	×	×	×
Agents						
OpenHands* [6]	×	✓	✓	✓	✓	✓
Devin [5]	×	✓	✓	✓	✓	✓
Claude Code [7]	×	✓	✓	✓	✓	✓
Cline [50]	×	✓	✓	✓	✓	✓
Replit [51]	×	✓	✓	×	✓	✓

Table 1: **Comparison of popular AI coding copilots and agents**, where ✓ denotes the action is done by the AI, ✗ denotes the action is approved by a human, and × denotes the feature does not exist. We split features based on basic ones that involve the current file and more advanced capabilities. We use * to denote the choice of copilot and agent that we select for this study.

Task Design

1. Data Analysis
 - a. 2 problems in GAIA
 - b. Extend the question to include additional instructions about visualizing the curated data
2. Feature Addition
 - a. 2 problems based on SWE-bench
 - b. Add features to matplotlib
3. Bug Fixing
 - a. 2 problems in GAIA
 - b. Challenging and potentially benefits from having a human-in-the-loop

RQ1: Productivity

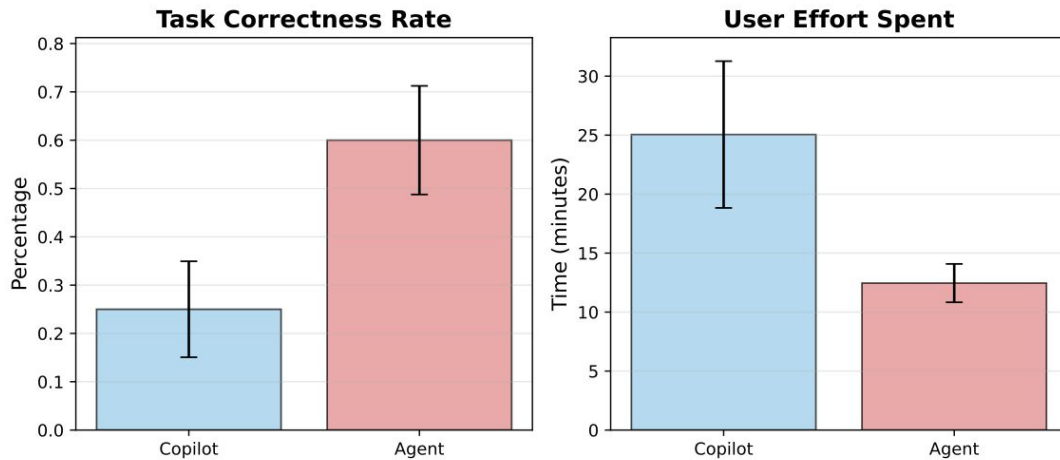


Figure 4: **Measuring the effect of different coding assistants on user productivity.** We consider two measurements: the task correctness rate and user effort spent to build a correct solution (i.e., excluding incomplete or incorrect solutions). For each metric, we report the mean and standard error.

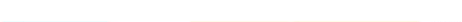
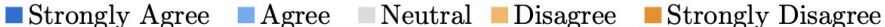
RQ1: Productivity

RQ1 Findings

Compared to copilots, agents improve the task correctness rate and reduce the effort required for users to complete the task. New workflows that encourage multi-tasking are needed to most effectively leverage agents.

RQ2: User Experiences

Table 2: Comparing user experience with Github Copilot and OpenHands. We report user rating distributions of Likert scale responses (ranging from strongly agree to strongly disagree) to various statements. We order the questions based on user preferences for OpenHands versus GitHub Copilot.

Comparison Statement	Distribution
C1 I experienced less cognitive load using OpenHands than Github Copilot	75%  10%
C2 I was able to accomplish new tasks with OpenHands than Github Copilot	70%  5%
C3 I was able to accomplish tasks faster with OpenHands than Github Copilot	50%  35%
C4 I was more in the flow using OpenHands than Github Copilot	45%  30%
C5 I was more satisfied using OpenHands than Github Copilot	45%  40%
C6 I have a better understanding of OpenHands outputs than Github Copilot outputs	25%  55%
	

RQ2: User Experiences

RQ2 Findings

Agents were significantly preferred over copilots in only 2 out of 6 comparisons, indicating room for improving user experience in human-agent collaboration.

RQ3: Interaction Comparisons

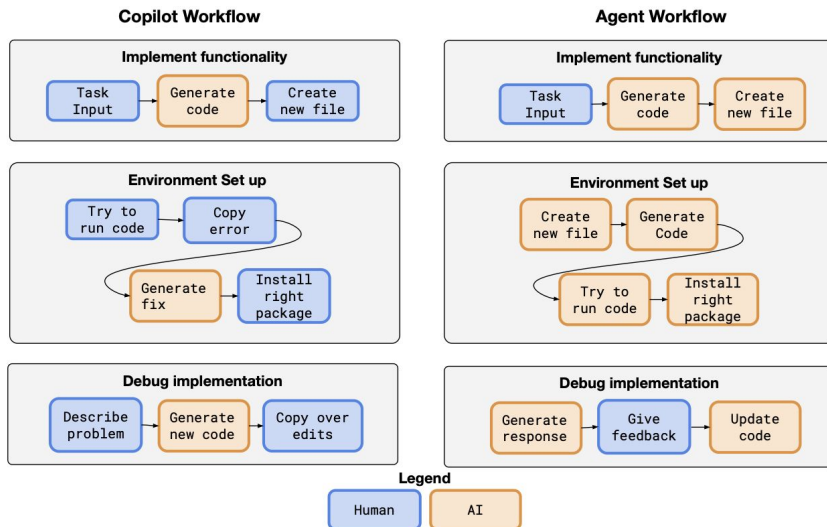


Figure 5: **Comparing developer workflows with copilots and agents.** We visualize abstracted user trajectories across three parts of the workflow: implementing functionality, setting up the environment, and debugging. We find that agentic workflows begin to shift more effort to the AI, aligning with the observed time saved.

RQ3: Interaction Comparisons

RQ3 Findings

Agentic workflows are more AI-driven than copilot workflows, where human effort is largely used to oversee agent actions in the debugging phase.

Design Desiderata

Table 3: Post-study survey responses about future usage of copilots compared to agents on the Likert scale.

Future Usage Statements	Distribution
F1 OpenHands is more competent than a human programmer	70%  10%
F2 Moving forward, I would continue to use OpenHands over Github Copilot	20%  60%
■ Strongly Agree ■ Agree ■ Neutral ■ Disagree ■ Strongly Disagree	

Design Desiderata

1. Agent behaviors should be transparent to users.
2. Agents should balance the level of proactivity.
3. Agent should effectively leverage human effort.

Final Conclusion

- Across scales, a consistent picture emerges:
 - **Macro:** AI work is spreading but uneven.
 - **Meso:** Coordination and evaluation frameworks define new agent workplaces.
 - **Micro:** Human roles shift toward supervision and design.
- Together, these layers outline the emerging *Agent Economy* — one where firms, agents, and humans co-evolve.